

Kyaw Sithu

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Platform & DevOps Engineer | CNCF Kubestronaut | kst-devops.com | github.com/rosaliei

Platform and DevOps engineer with 9+ years across fintech, telecom, healthcare and ISP environments, including 4+ years running production Kubernetes on Amazon EKS and on-premises RKE2. CNCF Kubestronaut holding all five Kubernetes certifications, plus ICA, AWS SAA-03 and RHCSA. Cut AWS spend by 20% and release lead time from 1 day to under 30 minutes for production payment platforms. Specialist in GitOps delivery, Istio service mesh and zero-trust security with HashiCorp Vault, Boundary, Keycloak and Okta.

EXPERIENCE

DevOps Engineer

Apr 2022 - Present

Peg B Technology, Dubai, UAE

Promoted from System Engineer to DevOps Engineer in 2024; now lead platform engineering for regulated payment workloads.

- Operate the payment gateway and application platform serving fintech and telecom customers: 40+ microservices across 6 EKS and RKE2 clusters.
- Launched Whizmo payment wallets in the UAE and Costa Rica, leading incident management from go-live to steady operations across 2 regions.
- Implemented GitOps with Jenkins, GitHub Actions and Kustomize, cutting release time from about a day to under 30 minutes.
- Reduced AWS spend by 20% via EKS right-sizing, Spot non-production, S3 lifecycle policies and idle-environment shutdown.
- Maintained Kafka disaster recovery with tested primary-to-DR and DR-to-primary switchovers completed in under 30 minutes.
- Standardized secrets on HashiCorp Vault and HCP for 40+ microservices, removing static credentials, with Keycloak OIDC behind API Gateway.
- Engineered observability with Prometheus, Grafana and ELK (Fleet, APM) across all 6 clusters for faster troubleshooting and root-cause analysis.
- Managed Istio in production with mutual TLS and ingress/egress policy, encrypting service-to-service traffic across clusters.
- Introduced AI coding agents and MCP integrations for Terraform, manifests and runbooks, removing repetitive manual editing from daily work.
- Produced capacity forecasts and 20+ technical runbooks for cluster scaling, failover and onboarding, cutting onboarding time by 50%.
- Automated AWS provisioning (SSM, S3, KMS, cross-account IAM) with Terraform, Ansible and Packer, cutting environment builds from 2-3 days to 2-3 hours.

System Engineer

Jul 2020 - Apr 2022

GTMH Telecom, Myanmar

- Built and maintained VMware ESXi private cloud hosting 80+ VMs for telecom and fintech services.
- Introduced Docker containerization and early Kubernetes workloads, replacing 30+ bare-metal deployments.
- Automated provisioning of 50+ servers with Ansible and Bash, reducing build time from days to hours.
- Ran Zabbix and ELK monitoring for 80+ VMs, hardened Linux images and managed OS lifecycle.

System Engineer

Jul 2019 - Jul 2020

Fortune Broadband, Myanmar

- Operated ISP network and Linux infrastructure serving 10,000+ broadband subscribers, covering configuration, patching and core-service support.

System Support Engineer

Jan 2018 - Jan 2019

Sysnet System Pte Ltd, Myanmar

- Deployed hospital systems for 5+ healthcare clients, managing Windows/Linux servers, backups and user access.

Junior System Administrator

Dec 2016 - Jan 2018

RITZ Public Company Limited, Myanmar

- Administered Linux systems and networking; wrote first automation shell scripts for repetitive tasks.

PROJECTS

Zero-Trust Kubernetes Access & Monitoring

github.com/rosaliei/boundary-eks-vault-okta

Okta, HCP Boundary, Vault, Amazon EKS

- Engineered zero-trust access to a private EKS cluster with no VPN, no kubeconfig files and no standing credentials.
- Combined Okta SSO, HCP Boundary and Vault-issued 15-minute Kubernetes credentials mapped to RBAC, with auto-scaling Boundary workers.
- Automated the full stack with Terraform, Packer and Ansible; GitHub Actions deployed to AWS through OIDC with no static cloud keys.
- Monitored access sessions, credential issuance and cluster health with Prometheus and Grafana.

GitOps Platform Monorepo

github.com/rosaliei/gitops-monorepo

EKS, Argo CD, Prometheus, Grafana, Karpenter, Terraform

- Created a single GitOps repository where Argo CD deploys applications and platform add-ons to EKS, with environment promotion by pull request.
- Covered cluster health, workload performance and compute cost with Prometheus and Grafana dashboards and alerts.
- Cut compute cost with Karpenter Spot instances and node consolidation; provisioned all infrastructure with Terraform.

MCP Server for Kubernetes

github.com/rosaliei/mcp-k8s

MCP, Kubernetes API, Python

- Developed a Python-based Model Context Protocol (MCP) server letting AI agents safely query a Kubernetes cluster: workloads, resources, logs and events.
- Designed read-only access under a dedicated service account scoped by Kubernetes RBAC, so an agent could investigate but never change the cluster.

Keycloak SSO Federation Gateway

github.com/rosaliei/keycloak-ssso-federation-gateway

Keycloak, NGINX/njs, JavaScript

- Wrote a JavaScript (njs) API gateway letting a web application keep a custom login UI while Keycloak owns identity and credentials.
- Traced and documented 7 authentication flows end to end with architecture diagrams.

CERTIFICATIONS (credly.com/users/kst1996)

- CNCF Kubestronaut: CKA, CKAD, CKS, KCNA, KCSA
- AWS Solution Architect Associate
- Istio Certified Associate (ICA)
- Red Hat Certified System Administrator (RHCSA)

EDUCATION

Bachelor of Computing, Computer Science pathway

Expected 2027

NCC Education / Strategy First University

TECHNICAL WRITING & COMMUNITY

Technical Writer

Present

kst-devops.com

- Publish a technical blog on zero-trust access across AWS, Azure and GCP, Entra ID SSO and Keycloak federation, with 30+ architecture diagrams and runbooks.

Moderator

2022 - Present

Dubai IT Community

- Guide job seekers and share hands-on cloud, Kubernetes and DevOps knowledge with IT professionals in Dubai.

TECHNICAL SKILLS

- Cloud & Kubernetes: AWS (EKS, EC2, VPC, IAM, SSM, S3, KMS), RKE2/Rancher, Helm, Kustomize, Docker, VMware ESXi, Huawei Cloud
- IaC & CI/CD: Terraform, Ansible, Packer, Bash, GitHub Actions, Jenkins, GitOps, Git, Gitea, Bitbucket
- Security & Identity: HashiCorp Vault, Boundary, Keycloak, Okta, Microsoft Entra ID, OIDC/SSO, Linux hardening
- Service Mesh & Gateways: Istio (sidecar, ambient), mutual TLS, Consul, NGINX, Kong, VPC peering
- Observability & Data: Prometheus, Grafana, ELK (Fleet, APM), Datadog, Zabbix, LibreNMS, Kafka, Kafka Connect
- Development & AI: Python, JavaScript (Node.js, njs), REST APIs, OIDC/JWT, AI coding agents, MCP